

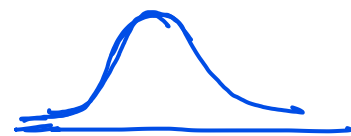
$$X \sim \text{Exp}(\lambda)$$

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$P(X > 2) = \int_2^{\infty} \lambda e^{-\lambda x} dx =$$

3.5 PROBABILITY AND MOMENT INEQUALITIES

Let X be a random variable defined on a probability space $(\Omega, \mathcal{F}, P)$, and let $B \in \mathcal{B}_1$ be a Borel set. In many situation $P(\{X \in B\})$ cannot be explicitly evaluated and therefore some estimate of this probability may be desired. For example if a random variable Z has the p.d.f.

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}, -\infty < z < \infty,$$


then

$$P(\{Z > 2\}) = \int_2^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz \quad (5.1)$$

cannot be explicitly evaluated and, therefore, an estimate of this probability may be desired. To estimate this probability one has to either resort to numerical integration or use some other estimation procedure. Inequalities are popular estimation procedures and they play an important role in probability theory.

Let X be a random variable.

3.5.1 Markov Inequality

$$r=1$$

$$E(X)$$

Suppose that $E(|X|^r) < \infty$, for some $r > 0$. Then, for any $c > 0$,

$$P(\{|X| \geq c\}) \leq \frac{E(|X|^r)}{c^r}.$$

$$P(|X| \geq c) \leq \frac{E(|X|)}{c}$$

• Let X is a nonnegative RV

$$P(X \geq c) \leq \frac{E(X)}{c}$$

3.5.2 Chebyshev Inequality

Suppose that X has finite first two moments. If $\mu = E(X)$ and $\sigma^2 = \text{Var}(X)$ ($\sigma \geq 0$). Then for any $k > 0$,

$$P(\{|X - \mu| \geq k\}) \leq \frac{\sigma^2}{k^2}.$$

$$\Rightarrow 1 - P(|X - \mu| < k) \leq \frac{\sigma^2}{k^2}$$

$$\Rightarrow \boxed{P(|X - \mu| < k) \geq 1 - \frac{\sigma^2}{k^2}}$$

Take $k = k'\sigma$

$$\Rightarrow P(|X - \mu| < k'\sigma) \geq 1 - \frac{\sigma^2}{k'^2 \sigma^2}$$

$$\Rightarrow \boxed{P(|X - \mu| < k'\sigma) \geq 1 - \frac{1}{k'^2}}$$

$$P(8 \leq X \leq 28) = P(8 - \mu \leq X - \mu \leq 28 - \mu)$$

$$= P(-10 \leq X - 18 \leq 10)$$

$$= P(|X - 18| \leq 10) \geq 1 - \frac{\sigma^2}{100} = 1 - \frac{6.25}{100}$$

$$= \frac{15}{16}$$

Ex: The number of customers who visit a store everyday is a R.V. with $\mu = 18$ and $\sigma = 2.5$. With what probability can we assert that there will be between 8 to 28 customers?

Example: 5.1

Let us revisit the problem of estimating $P(\{Z > 2\})$, defined by (5.1). Using Example 4.2 (iii), we have $\mu = E(Z) = 0$ and $\sigma^2 = \text{Var}(Z) = 1$. Moreover, using Example 4.2 (ii), $Z \stackrel{d}{=} -Z$. Consequently $P(\{Z > 2\}) = P(\{-Z > 2\}) (= P(\{Z < -2\}))$, i.e.,

$$P(\{Z > 2\}) = \frac{P(\{|Z| > 2\})}{2} = \frac{P(\{|Z| \geq 2\})}{2}.$$

Now, using the Chebyshev inequality we have

$$P(\{Z > 2\}) = \frac{P(\{|Z| \geq 2\})}{2} \leq \frac{E(|Z|^2)}{8} = \frac{1}{8} = 0.125.$$

The exact value of $P(\{Z > 2\})$, obtained using numerical integration, is 0.0228. ■

The following example illustrates that bounds provided in Theorem 5.1 and Corollary 5.1 are tight, i.e., the upper bounds provided there in may be attained.

Example 5.2

✓ Let X be a random variable with p.m.f.

$$f_X(x) = \begin{cases} \frac{1}{8}, & \text{if } x \in \{-1, 1\} \\ \frac{3}{4}, & \text{if } x = 0 \\ 0, & \text{otherwise} \end{cases}.$$

Clearly $E(X^2) = \frac{1}{4}$ and, therefore, using the Markov inequality we have

$$\underline{P(\{|X| \geq 1\})} \leq E(X^2) = \frac{1}{4}.$$

The exact probability is

$$P(\{|X| \geq 1\}) = P(\{X \in \{-1, 1\}\}) = \frac{1}{4}. \blacksquare$$

Example 2.49 Suppose we know that the number of items produced in a factory during a week is a random variable with mean 500.

- ✓(a) What can be said about the probability that this week's production will be at least 1000?
- ✓(b) If the variance of a week's production is known to equal 100, then what can be said about the probability that this week's production will be between 400 and 600?

Solution: Let X be the number of items that will be produced in a week.

(a) By Markov's inequality,

$$P\{X \geq 1000\} \leq \frac{E[X]}{1000} = \frac{500}{1000} = \frac{1}{2}$$

(b) By Chebyshev's inequality,

$$P\{|X - 500| \geq 100\} \leq \frac{\sigma^2}{(100)^2} = \frac{1}{100}$$

Hence,

$$P\{|X - 500| < 100\} \geq 1 - \frac{1}{100} = \frac{99}{100}$$

and so the probability that this week's production will be between 400 and 600 is at least 0.99. ■

